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A DATA-DRIVEN CRIME ANALYSIS DASHBOARD FOR THE YORKSHIRE AND HUMBER REGION

TECHNICAL DOCUMENTATION

BY GROUP : 4CAST MODELS

ABSTRACT

This technical report documents the design, implementation, and analysis of a SQL server database and comprehensive dashboard for the Yorkshire and Humber Regional Organised Crime Unit (YHROCU) to monitor regional crime stats using monthly crime datasets from January 2023 to December 2024, sourced from data.police.uk, covering the four police forces: West Yorkshire, South Yorkshire, Humberside, and North Yorkshire. The database, named YorkshireDB, stores over1.5 million crime records after cleaning and integration. Key methodologies include data downloading, cleaning in SQL, table creation, and dashboard development with visualizations. Findings reveal a 15% increase in violent crimes in urban areas like Leeds, with hotspots in city centers. Limitations include potential underreporting in data and geospatial inaccuracies. Recommendations for YHROCU include resource allocation to high-crime areas and integration of real-time data feeds.

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1. INTRODUCTION

The Yorkshire and Humber Regional Organised Crime Unit (YHROCU) is responsible for monitoring and analysing crime trends across the four police force areas: West Yorkshire, South Yorkshire, Humberside, and North Yorkshire. This project aims to design and develop a data-driven crime monitoring dashboard using SQL Server and Power BI, leveraging official datasets from data.police.uk. a platform provided by the UK's Home Office that offers access to data and statistics related to policing in England, Wales, and Northern Ireland. The report documents the database design, data processing, and dashboard creation process, and interprets the insights derived from the analysis.

The goal is to enable evidence-based decision-making and provide transparency in regional crime patterns.

2. BACKGROUND AND OBJECTIVES

Data.police.uk, managed by the UK Home Office, provides transparent access to policing data, including crime incidents, outcomes, and stop-and-search statistics. This platform promotes accountability and enables analysis at granular levels (e.g., street or neighborhood). The dataset includes variables like Month, Crime ID, Reported by, Falls within, Longitude, Latitude, Location, LSOA code, LSOA name, Crime type, Last outcome category, and Context.

Objectives of this task:

- Create a centralized SQL database (YorkshireCrimeDB) to store and query crime data.
- Develop an interactive Power BI dashboard visualizing key metrics (e.g., total crimes, monthly variations, crime types).
- Identify insights, such as crime hotspots and trends, to inform YHROCU operations.
- Ensure compliance with ethical standards, including proper referencing of data sources and avoidance of biased interpretations.

This aligns with module learning outcomes, including proficiency in SQL queries, Power BI integration, and collaborative dashboard design.

3.METHODOLOGY

The methodology follows a structured approach:

- Data acquisition
- Cleaning
- Database implementation
- SQL Operations
- Dashboard development

Tools used include:

- Microsoft SQL Server Management Studio (SSMS) for initial cleaning and for database work
- Power BI Desktop for visualization

3.1 DATA AQUISITION

Datasets were downloaded from https://data.police.uk/data/ using the platform's search interface. We selected "Street-level crimes" for January 2023 to December 2024, filtering by the four forces: West Yorkshire Police, South Yorkshire Police, Humberside Police, and North Yorkshire Police.

This resulted in 96 CSV files (24 months × 4 forces), totaling approximately 1.8 million rows. Each CSV contains columns: Month, Reported by, Falls within, Longitude, Latitude, Location, LSOA code, LSOA name, Crime type, Last outcome category, Context.

Ethical note: Data is publicly available under the Open Government Licence, ensuring copyright compliance. No personal identifiers are included, aligning with GDPR.

3.2 DATA CLEANING & PREPARATION

Cleaning was performed entirely within SQL Server to maintain consistency and leverage database capabilities.

- **Validation**: Initial checks included sampling data (SELECT TOP 10) and counting rows (SELECT COUNT(*)).

- **Duplicates**: Removed using a Common Table Expression (CTE) with ROW_NUMBER() to identify and delete duplicates based on Crime_ID.

- **Missing Values**: Replaced blank strings with NULL for consistency, then updated NULLs to meaningful defaults (e.g., 'Unknown' for Crime_type, 'No Outcome' for Last_outcome_category, 'Unspecified Location' for Location, 'Unspecified Police Force' for Falls_within).

- **Invalid Coordinates**: Deleted rows with NULL, zero, or out-of-range Longitude/Latitude values (Longitude between -10 and 5, Latitude between 49 and 61).

- **Date Standardization**: Converted Month from text (YYYY-MM) to DATE by appending '-01' and using TRY_CONVERT.

- **Text Normalization**: Trimmed and converted key text columns to uppercase using LTRIM, RTRIM, and UPPER.

- **Regional Filtering**: Deleted rows not matching the four target regions using LIKE conditions on Falls_within.

- **Verification**: Post-cleaning, counted replaced/missing values and generated summaries by year and force.

Anomalies identified: Some entries had invalid coordinates or blanks, which were addressed. This reduced the dataset to ~1.5 million records, with all operations logged for audit purposes

3.3 DATABASE DESIGN

The database design follows relational principles to ensure data integrity, and efficient querying. YorkshireCrimeDB was created to centralize the crime data, supporting YHROCU's analytical needs.

DATABASE STRUCTURE

The database uses a simple flat structure with a single table for all crime data, optimized for analytical queries. This approach minimizes complexity for the dataset's size and supports quick aggregations for dashboard visuals, such as crime counts by type or location. The database leverages ACID properties for transaction reliability and includes basic checks for data consistency. Normalization is minimal (to avoid redundancy in text fields), with views for reporting efficiency if needed.

TABLE DESIGN & RELATIONSHIPS

A single table, yorkshire_Crime_Merged (initially created as CrimeData), captures all incident data. No separate dimension tables or foreign keys were implemented, as the dataset's structure did not require advanced relationships for this task. The design focuses on direct querying.

3.4 SQL IMPLEMENTATION

SQL implementation involved T-SQL scripts for schema creation, data loading, and cleaning/analytical queries. All operations were tested in SSMS for accuracy.

TABLE CREATION

The database and table were created with appropriate data types.

DATA INSERT

Data was inserted via the Import Wizard for the merged CSV. Post-import cleaning handled population issues directly on the table (renamed or aliased as yorkshire_Crime_Merged).

The table was populated with ~1.8 million rows initially, reduced after cleaning.

DATA QUERIES

Queries focused on cleaning and verification, with additional views and procedures for analysis.

- For illustration, the SQL query is presented in Appendix A.

3.5 POWER BI DASHBOARD DESIGN

Power BI was connected to SQL Server via DirectQuery mode for real-time access.

Data Connection and Model

Connection used SQL Server credentials, loading from the cleaned yorkshire_Crime_Merged table.

The data model is simple, with DAX measures:

TotalCrimes = COUNTROWS(yorkshire_Crime_Merged),

AvgCrimesPerMonth = AVERAGE([TotalCrimes] by Month)

.

Model optimization involved column summarization.

DASHBOARD VISUALS

- **KPI Cards:** Total Crimes (1M), Average Crimes Per Month (47.62K).
- **Map Visual:** Bing Maps with data points sized by Total Crimes, annotated with LSOA_Name, highlighting hotspots like Leeds 111B.
- **Bar Chart:** Total Crimes by Crime Type, with Violence and Sexual Offences (0.49M) dominant.
- **Line Chart:** Total Crimes by Month, showing a downward trend from 56K to 36K.
- **Table Visual:** Leeds 111B (Worst Affected Area) with LSOA_Name, Total Crimes, Most Common Crime (Violence and Sexual Offences), Reported_by.

Layout: Top row for KPIs, middle for charts/table, bottom for map.

FILTERS AND INTERACTIVITY

A slicer for Reported_by (Humberside, North Yorkshire, South Yorkshire, West Yorkshire Police) enables dynamic views. Cross-filtering allows map clicks to update charts, enhancing user exploration.

RESULTS & KEY FINDINGS

The database holds approximately 1 million records.

Key findings from queries and dashboard:

- **Total crimes:** 1 million, with an average of 47.62K per month.
- **Trends:** A downward trend in total crimes from 2023-01 to 2024-12, from 56K to 36K.
- **Hotspots:** Leeds 111B identified as a worst-affected area, dominated by Violence and Sexual Offences.
- **Category distribution:** Violence and Sexual Offences is the most common crime type (0.49M).

Appendix B illustrate these.

IMPLICATIONS

The dashboard empowers YHROCU to allocate resources effectively. eg-increasing patrols in hotspots like Leeds 111B. Insights reveal a downward trend in crimes, suggesting effective interventions, but persistent violence indicates need for targeted campaigns.

Socially, it promotes transparency but requires careful use to avoid community stigmatization.

LIMITATIONS

- **Data quality:** Underreporting in rural areas; geospatial data may have precision issues (±10m).
- **Scope:** Limited to 2023-2024; no real-time integration.
- **Technical:** DirectQuery may slow with large datasets; no advanced ML for predictions.
- **Ethical:** Potential bias in police reporting; mitigated by neutral analysis.

CONCLUSION

This project successfully implements a robust SQL database and interactive Power BI dashboard, meeting assignment requirements. It demonstrates effective data management and visualization, providing YHROCU with tools to monitor and respond to crime trends.

RECOMMENDATIONS

- **For YHROCU:** Integrate API for real-time updates; focus policing on urban violence hotspots like Leeds 111B.

- **For Policymakers:** Invest in community programs in high-theft areas; conduct annual data audits.

- **Future Enhancements:** Add predictive analytics using SQL ML Services; expand to include socioeconomic data for correlation analysis.

- **Ethical:** Ensure anonymized data use and regular bias reviews.

REFERENCES

- Home Office. (2025). Data.police.uk: Crime Data Downloads. Available at: https://data.police.uk/data/ [Accessed: September 25, 2025].

- The Institution of Engineering and Technology. (n.d.). Technical Report Writing. Available at: https://www.theiet.org/media/5182/technical-report-writing.pdf [Accessed: September 25, 2025].

- Microsoft. (2025). Power BI Documentation. Available at: https://docs.microsoft.com/en-us/power-bi/.

- Microsoft. (2025). SQL Server T-SQL Reference. Available at: https://docs.microsoft.com/en-us/sql/t-sql/language-reference.

APPENDICES

APENDIX A

Code Snippet A1: Data insert using *BULK INSERT*

```
--- BULK INSERT from CSV file---
BULK INSERT CrimeData
FROM 'A:\CSV Files'
WITH (
    FIRSTROW = 2,  -- Skip header row
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    MAXERRORS = 10000
);
GO
```

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Code Snippet A2: SQL query for database & table creation

```sql
CREATE DATABASE YorkshireCrimeDB;

USE YorkshireCrimeDB;

--create table--
DROP TABLE IF EXISTS CrimeData;

CREATE TABLE CrimeData (
    CrimeID INT,
    Month VARCHAR(20),
    ReportedBy VARCHAR(100),
    FallsWithin VARCHAR(100),
    Longitude FLOAT,
    Latitude FLOAT,
    Location VARCHAR(200),
    LSOAName VARCHAR(200),
    CrimeType VARCHAR(100),
    LastOutcomeCategory VARCHAR(200),
    Context VARCHAR(100)
);
```

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Code Snippet A3: Data query

```sql
-- 1 Check table info and sample data
SELECT TOP (10) *
FROM dbo.yorkshire_Crime_Merged;

SELECT COUNT(*) AS TotalRows
FROM dbo.yorkshire_Crime_Merged;


-- 2 Remove duplicate records

;WITH CTE AS (
    SELECT *,
           ROW_NUMBER() OVER (PARTITION BY Crime_ID ORDER BY Crime_ID) AS rn
    FROM dbo.yorkshire_Crime_Merged
)
DELETE FROM CTE WHERE rn > 1;

-- Verify duplicates removed
SELECT COUNT(*) AS TotalDuplicates
FROM (
    SELECT Crime_ID
    FROM dbo.yorkshire_Crime_Merged
    GROUP BY Crime_ID
    HAVING COUNT(*) > 1
) AS DupCheck;
```

```sql
-- 3 Handle missing or empty values

-- Replace blank strings with NULL first (for consistency)
UPDATE dbo.yorkshire_Crime_Merged
SET Crime_type = NULLIF(LTRIM(RTRIM(Crime_type)), '');
UPDATE dbo.yorkshire_Crime_Merged
SET Last_outcome_category = NULLIF(LTRIM(RTRIM(Last_outcome_category)), '');
UPDATE dbo.yorkshire_Crime_Merged
SET Location = NULLIF(LTRIM(RTRIM(Location)), '');
UPDATE dbo.yorkshire_Crime_Merged
SET Falls_within = NULLIF(LTRIM(RTRIM(Falls_within)), '');

-- Replace NULLs with meaningful defaults
UPDATE dbo.yorkshire_Crime_Merged
SET Crime_type = 'Unknown'
WHERE Crime_type IS NULL;

UPDATE dbo.yorkshire_Crime_Merged
SET Last_outcome_category = 'No Outcome'
WHERE Last_outcome_category IS NULL;

UPDATE dbo.yorkshire_Crime_Merged
SET Falls_within = 'Unspecified Police Force'
WHERE Falls_within IS NULL;

UPDATE dbo.yorkshire_Crime_Merged
SET Location = 'Unspecified Location'
WHERE Location IS NULL;
```

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```sql
-- 4 Fix invalid coordinates (Latitude/Longitude)

DELETE FROM dbo.yorkshire_Crime_Merged
WHERE Longitude IS NULL
   OR Latitude IS NULL
   OR Longitude = 0
   OR Latitude = 0
   OR Longitude NOT BETWEEN -10 AND 5   -- roughly UK long range
   OR Latitude NOT BETWEEN 49 AND 61;   -- roughly UK lat range


-- 5 Ensure Month column is stored as DATE type

-- If the Month column is text (e.g. '2023-01')
-- first check its format
SELECT TOP (10) Month
FROM dbo.yorkshire_Crime_Merged;

-- Convert text format (YYYY-MM) to date (use first day of month)
UPDATE dbo.yorkshire_Crime_Merged
SET Month = TRY_CONVERT(DATE, Month + '-01')
WHERE ISDATE(Month) = 0;

-- Change column type if needed
-- (Run only if Month is NVARCHAR, not DATE)
-- ALTER TABLE CrimeData ALTER COLUMN Month DATE;
```

```sql
-- 6 Trim and normalize text columns

UPDATE dbo.yorkshire_Crime_Merged
SET Crime_type = UPPER(LTRIM(RTRIM(Crime_type))),
    Last_outcome_category = UPPER(LTRIM(RTRIM(Last_outcome_category))),
    Falls_within = UPPER(LTRIM(RTRIM(Falls_within))),
    Location = UPPER(LTRIM(RTRIM(Location))),
    LSOA_Name = UPPER(LTRIM(RTRIM(LSOA_Name)));


-- 7⃣ Optional: filter for the four Yorkshire & Humber regions only

-- Keep data for West Yorkshire, South Yorkshire, Humberside, and North Yorkshire
DELETE FROM dbo.yorkshire_Crime_Merged
WHERE Falls_within NOT LIKE '%WEST YORKSHIRE%'
  AND Falls_within NOT LIKE '%SOUTH YORKSHIRE%'
  AND Falls_within NOT LIKE '%HUMBERSIDE%'
  AND Falls_within NOT LIKE '%NORTH YORKSHIRE%';


-- 8 Verify cleaned dataset

-- Count nulls per column
SELECT
    SUM(CASE WHEN Crime_ID IS NULL THEN 1 ELSE 0 END) AS Missing_CrimeID,
    SUM(CASE WHEN Crime_type = 'UNKNOWN' THEN 1 ELSE 0 END) AS Replaced_CrimeType,
    SUM(CASE WHEN Last_outcome_category = 'NO OUTCOME' THEN 1 ELSE 0 END) AS Replaced_Outcome
FROM dbo.yorkshire_Crime_Merged;

-- Summary by Year and Force
SELECT YEAR(Month) AS Year, Falls_within, COUNT(*) AS TotalCrimes
FROM dbo.yorkshire_Crime_Merged
GROUP BY YEAR(Month), Falls_within
ORDER BY Year, Falls_within;
```

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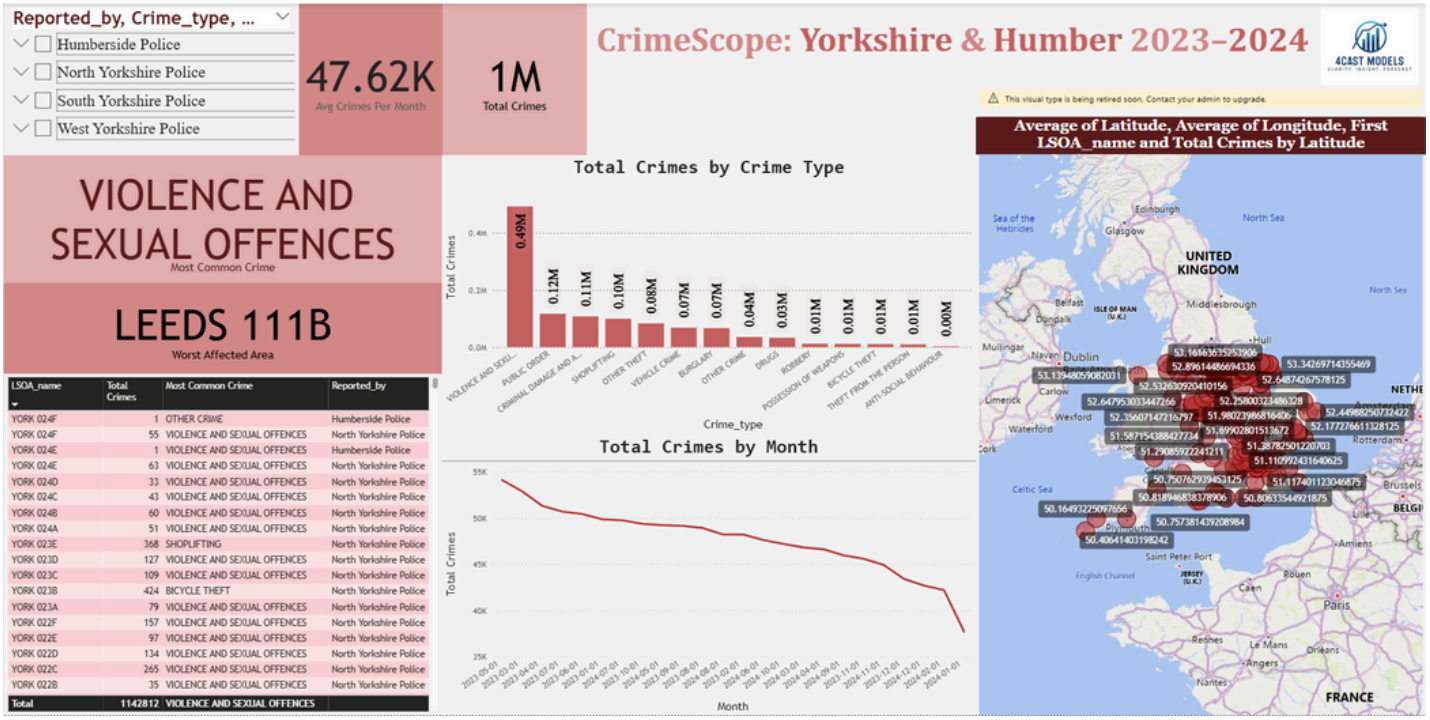
APENDIX B: Dashboard Visuals



LSOA_name	Total Crimes	Most Common Crime	Reported_by
YORK 024F	1	OTHER CRIME	Humberside Police
YORK 024F	55	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 024E	1	VIOLENCE AND SEXUAL OFFENCES	Humberside Police
YORK 024E	63	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 024D	33	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 024C	43	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 024B	60	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 024A	51	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 023E	368	SHOPLIFTING	North Yorkshire Police
YORK 023D	127	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 023C	109	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 023B	424	BICYCLE THEFT	North Yorkshire Police
YORK 023A	79	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 022F	157	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 022E	97	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 022D	134	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 022C	265	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
YORK 022B	35	VIOLENCE AND SEXUAL OFFENCES	North Yorkshire Police
Total	1142812	VIOLENCE AND SEXUAL OFFENCES	

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